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Using Public Health Detailing to Increase Access and Confidence in COVID-19 Vaccines and Reinvest in Disproportionately Impacted NYC Communities

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ABSTRACT

Context: COVID-19 vaccination rates in New York City (NYC) began to plateau in the spring of 2021, with unacceptable inequities in vaccination rates based on race.

Program: To address racial inequities in vaccination rates and COVID-19 health outcomes, the New York City Department of Health and Mental Hygiene adapted a preexisting provider outreach and education program for public health emergency use with the goals of community reinvestment and increasing patient confidence and access to the COVID-19 vaccines. The Vaccine Public Health Detailing (VPHD) program was delivered as part of a larger community outreach initiative and brought timely updates, materials, and access to technical assistance to primary care providers and staff in NYC neighborhoods experiencing COVID-19 health inequities. Outreach representatives also collected feedback from providers on resource needs to inform the agency's response.

Implementation: Sixteen outreach representatives were rapidly trained on COVID-19–related content and strategic communication techniques and launched a 3-wave campaign across targeted neighborhoods throughout NYC. The campaign ran from May 2021 to March 2022 and was conducted in coordination with other community engagement initiatives aimed at the general public to promote greater collective impact.

Evaluation: In total, 2873 detailing sessions were conducted with 2027 unique providers at 1281 sites. Outreach representatives successfully completed visits at more than 85% of practices that were in scope and operating. Approximately 20% (285) of the sites requested a referral for technical assistance to become a COVID-19 vaccination site or enroll in the Citywide Immunization Registry. Qualitative information shared by providers offered a more in-depth understanding of vaccine-related sentiments and challenges faced by providers on the ground.

Discussion: VPHD is an effective method for supporting community providers, gathering feedback on resource needs and practice challenges, and increasing health systems efficacy during a public health emergency while also prioritizing racial equity and community reinvestment.

KEY WORDS: community outreach, COVID-19 vaccination, detailing, health equity, provider education, provider outreach

Context

In mid-2021, COVID-19 vaccination rates in New York City (NYC) began to plateau, with significant inequities persisting by race.[1] To increase patient confidence and access to COVID-19 vaccines, the New York City Department of Health and Mental Hygiene (DOHMH) rapidly adapted its Public Health Detailing (PHD) program to deliver timely education and tools to primary care providers in the Taskforce on Racial Inclusion and Equity (TRIE) neighborhoods,

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which were established in response to the disproportionate impact of COVID-19 on communities of color.[2,3] In July 2021, the time of full program launch, only 19% of TRIE neighborhoods were considered "fully vaccinated neighborhoods"—defined as 70% or more of adults residing in the neighborhood having a full initial vaccine course (1 or 2 doses, depending on the vaccine)—compared with 64% of non-TRIE neighborhoods.[4,5]

Program

PHD is a provider outreach and education program aimed at increasing provider knowledge and promoting preventive care practices and systemized care delivery.[6] Adapted from pharmaceutical detailing, which has its own complex history, PHD promotes health equity by using a place-based approach in the health topics selected and catchment areas for program delivery, which are informed by local data on health inequities. Campaigns consist of in-person visits by DOHMH outreach representatives (reps) who deliver evidence-based, topic-specific recommendations and "Action Kits" to providers and staff, which contain tools and resources to assist in implementing the campaign's recommendations into routine clinical practice.[7] Since the program's inception, PHD has conducted nearly 60 campaigns on diverse topics such as smoking cessation, hypertension, depression, and influenza.[8] The campaign described within, known as "Vaccine Public Health Detailing" (VPHD), expanded beyond PHD's typical catchment area of neighborhoods served by DOHMH's Bureaus of Neighborhood Health, formerly called District Public Health offices.[9]

VPHD operated as part of the Provider and Community Engagement (PACE) Response Group,[4] an integral part of DOHMH's emergency response structure. PACE advanced place-based community and provider engagement strategies to reduce racial and other health inequities in COVID-19 cases, hospitalizations, deaths, and vaccination through capacity building and partnership with community and faith-based organizations, community health workers, and medical providers and practices. Community outreach and provider engagement were conducted in tandem in target communities with the vision that these collective efforts would be greater than the sum of the individual program parts and lead to meaningful community reinvestment. VPHD also provided a crucial avenue for PACE to directly engage with and learn from community providers and disseminate COVID-19–related resources and information. Justification for VPHD was further bolstered by evidence suggesting a strong recommendation from a health care provider is associated with increased patient vaccine uptake.[10]

Implementation

Campaign development and strategy

Development of VPHD was an interdisciplinary process, leveraging expertise from across DOHMH including external affairs, clinical and vaccine operations, health systems support, provider and community engagement, and PHD leadership. Rather than developing written materials de novo for this campaign—a process that typically takes 12 to 16 months and is not well-suited to rapidly evolving messaging—PHD relied on existing agency resources to create the Action Kit.

In this 3-wave campaign, reps adopted elements of an e-detailing approach[11] and presented Action Kit materials on a tablet rather than in hard copy. This modification to the PHD approach, which typically utilizes hard copy materials to facilitate discussion between providers and reps, allowed reps to always have the latest and most up-to-date materials, which sometimes changed daily based on emerging scientific evidence.

As in all detailing campaigns,[7] the Action Kit was built around the campaign's key recommendations (Table). In addition to succinctly synthesizing the evidence, the key recommendations served to promote new initiatives such as the Vaccine Outreach and Counseling Program, a collaboration between DOHMH and selected payers that allowed providers to receive reimbursement for conducting outreach to unvaccinated patients.[12]

Materials were for both patient[13] and provider[14] audiences and included resources such as a provider communication guide to help overcome vaccine hesitancy and combat misinformation, as well as patient-facing infographics about how mRNA and adenovirus vaccines work (see Supplemental Digital Content 1, available at http://links.lww.com/JPHMP/B232, and Supplemental Digital Content 2, available at http://links.lww.com/JPHMP/B233). Patient materials were designed to be accessible to people with wide levels of health literacy and were available in at least the 13 languages most commonly spoken in NYC.

A distinguishing feature of VPHD relative to other outreach initiatives is that reps visited providers in person and directly discussed the provider's individual resource needs, thereby allowing reps to tailor their presentations and highlight the most relevant materials. Materials were presented as a "suite" of resources, with suggested pairings of how one material

TABLE
Campaign Key Recommendations

	Wave I (May 2021) and Wave II (Jun 2021-Oct 2021)	Wave III (Oct 2021-Mar 2022)
KR #1	Provide a clear, strong recommendation for COVID-19 vaccination at each clinical encounter to patients who are not yet vaccinated.	Provide a clear, strong recommendation for COVID-19 vaccination to all patients who are not yet vaccinated, including those with prior SARS-CoV-2 infection. Offer information on the efficacy, availability, and administration of the COVID-19 vaccines at every visit and document the encounter in the patient's medical record.
KR #2	Speak with patients about the COVID-19 vaccines and support informed decisions by providing honest answers to questions using plain language.	Reinforce that being fully vaccinated provides stronger and longer-lasting protection than natural immunity alone. Research demonstrates that antibody testing should not be used to guide decision-making around COVID-19 vaccination.
KR #3	Acknowledge any concerns patients might have, starting with empathy, followed by providing the facts.	Proactively outreach the need and availability of a COVID-19 third-dose vaccine for those who are immunocompromised or those patients eligible to receive a booster shot.
KR #4	Enroll in the Citywide Immunization Registry to identify patients not yet vaccinated.	Coadminister COVID-19 vaccine with routine immunizations to adults and adolescents. COVID-19 vaccines can be administered at the same time or at any time before or after other vaccines, including influenza vaccination.
KR #5	Offer vaccination (if possible) or help arrange vaccination for patients before they leave your office.[a]	Vaccinate on-site (if possible) or help arrange vaccination for patients before they leave your office.[a]

Abbreviations: KR, key recommendation; NYC, New York City.

[a] Guide patients in scheduling a vaccination appointment through NYC's COVID-19 Vaccine Finder or help make vaccination appointments for your patients by calling 877-VAX-4NYC (877-829-4692) and pressing 2 at the second prompt. You can also complete this form for patients to receive a call from the NYC Vaccine Appointment Hotline.

could be used with another to address particular barriers, which heightened the utility of the materials compared with viewing each alone. Reps e-mailed all providers within 24 hours of the visit, sharing links to the Action Kit materials. Sharing links rather than electronic attachments further ensured practices would have the latest materials, even if updates were made after the visit.

Staffing and management

The VPHD management team consisted of 7 members, all of whom work with the PHD program in similar roles in nonemergency times (1 director, 1 operations manager, 1 evaluator, an intern/program coordinator, and 3 outreach managers to oversee outreach reps). To scale up, VPHD partnered with the city's COVID-19 contact tracing program, NYC Health + Hospitals Test and Trace (T2). Through interagency collaboration, 16 community engagement specialists (CESs) were seconded to VPHD. CES staff were hired by T2 for the purpose of in-person engagement with hard-to-reach New Yorkers.[15] CESs, who typically did not have clinical training, represented diverse city communities and were able to speak at least one language in addition to English, skill sets that dovetailed with VPHD's needs.

Outreach managers oversaw a team of 3 to 5 CES reps. They conducted daily check-ins with their team members and accompanied CESs on selected visits for quality assurance and to ensure sound DOHMH representation. Managers elevated pressing questions or concerns to the director or operations manager, who would address the requests with agency content experts as needed. A weekly all-staff meeting served as a forum for the management team to proactively communicate any strategic or content changes and for reps to share experiences from the field.

Training

To prepare the CES staff for their role as outreach reps, the management team executed a comprehensive training covering COVID-19 subject matter, detailing techniques, and operational guidance and expectations.

First, staff received training on the detailing approach, including selling and communication strategies unique to PHD in the field of health care education. This included how to (1) frame the issue around the urgency to vaccinate and each provider's role in increasing population vaccination rates; (2) use compelling, local statistics from the practice's neighborhood—often presented at levels as granular as zip code to highlight racial and ethnic inequities and how they connect to historical disinvestment and policies rooted in systemic racism; (3) overcome barriers to implementing key recommendations at the

provider and practice levels, and helping providers understand how to overcome barriers with patients; (4) close the visit or "gain a commitment" from providers to utilize campaign materials in their practice.

Second, DOHMH subject matter experts delivered content presentations on the following: DOHMH's approach to equity; a technical overview of the COVID-19 vaccines, sufficient to allow reps to converse fluently with providers; treatment options including monoclonal antibodies; provider-patient communication techniques; strategies for promoting vaccination in youth and dispelling misinformation; and requirements for practices seeking to become vaccination sites.

Finally, staff were trained to administer a series of assessment questions at the start of each visit. Consistent with the detailing model, the assessment questions were used for 2 key purposes: to help reps tailor presentations to provider's needs and challenges and to collect self-reported data on current clinical practices that could inform agency planning.

Determining catchment area and data and reporting

VPHD completed 3 campaign waves, from May 2021 to March 2022, adapting and responding as needed to the changing COVID-19 environment. To determine the catchment area for detailing, the team considered both local data on vaccination rates and a racial equity framework, which included elements of the DOHMH's COVID-19 equity action plan.[16] The catchment area differed for each of the waves (Figure) and was determined as follows:

- *Wave I*: The first wave began with a smaller group of CESs and was treated as a pilot program to fine-tune operations. As such, wave I focused on 15 zip codes in the Bronx and Brooklyn that were part of the city's equity outreach plan; these areas were largely selected on the basis of low vaccination rates among Black residents. VPHD activities were concurrent with other community outreach activities in these areas.
- *Wave II*: Initial anecdotal feedback suggested VPHD was successful and well-received in its first wave; thus, DOHMH expanded VPHD to cover neighborhoods in all NYC boroughs in wave II. Specifically, inclusion criteria for wave II were as follows: neighborhoods with more than 20% Black residents and first-dose vaccination rate among adults at less than 60%; or neighborhoods with high COVID-19 mortality, defined as being in the top third of neighborhoods in NYC for fatality rates per 100 000 residents and less than 60% first-dose vaccination rates; or neighborhoods with first-dose vaccination rates at less than 60% that were included in the TRIE plan.[2,3] VPHD also strategically deployed outreach reps on an ad hoc basis to neighborhoods with surging case rates, such as Borough Park in Brooklyn and the South Shore of Staten Island. This dynamic, responsive approach was an important feature of the campaign and allowed for maximum impact where detailing was most needed.
- *Wave III*: VPHD revisited sites from the first and second waves to reinforce key messages and provide updated information and resources. In addition, practices in zip codes with a less than 70% first-dose vaccination rate that aligned with other DOHMH initiatives were included.

With respect to identifying practices, VPHD worked with agency colleagues to generate a list by utilizing DOHMH's internal records of medical practices as well as a purchased list. VPHD targeted any practice with at least one provider associated with the site in the following specialties: internal medicine; family practice; obstetrics/gynecology; and pediatrics. VPHD excluded hospital-based affiliates, as most NYC hospital systems had resources to develop standard referral protocols for central vaccination sites that made vaccines easily available to their patients.

VPHD collaborated with the agency's COVID-19 Interdisciplinary Data Team to build scalable data collection and reporting tools. For data collection, the team utilized Survey123, a product of ArcGIS. Survey123 enabled field-based data collection, and importantly, it featured location-based services that helped reps navigate unfamiliar territories across NYC by providing live updates on the closest assigned practice (as determined from a preloaded list of practices) so they could travel efficiently.

In addition, the team developed a daily-updated Tableau dashboard that utilized output from Survey123 to report on campaign reach and assessment question findings. This type of live reporting is atypical of detailing campaigns, as the frequently changing sample reflected a convenience sample and was not representative of the target neighborhoods as a whole; however, the team determined the value of immediate feedback superseded concerns around representativeness as the reports served as early warning signals for potential areas in need of further intervention. Rapid reports were triangulated alongside methodologically rigorous DOHMH surveys to confirm emerging trends and shifts in community feedback.

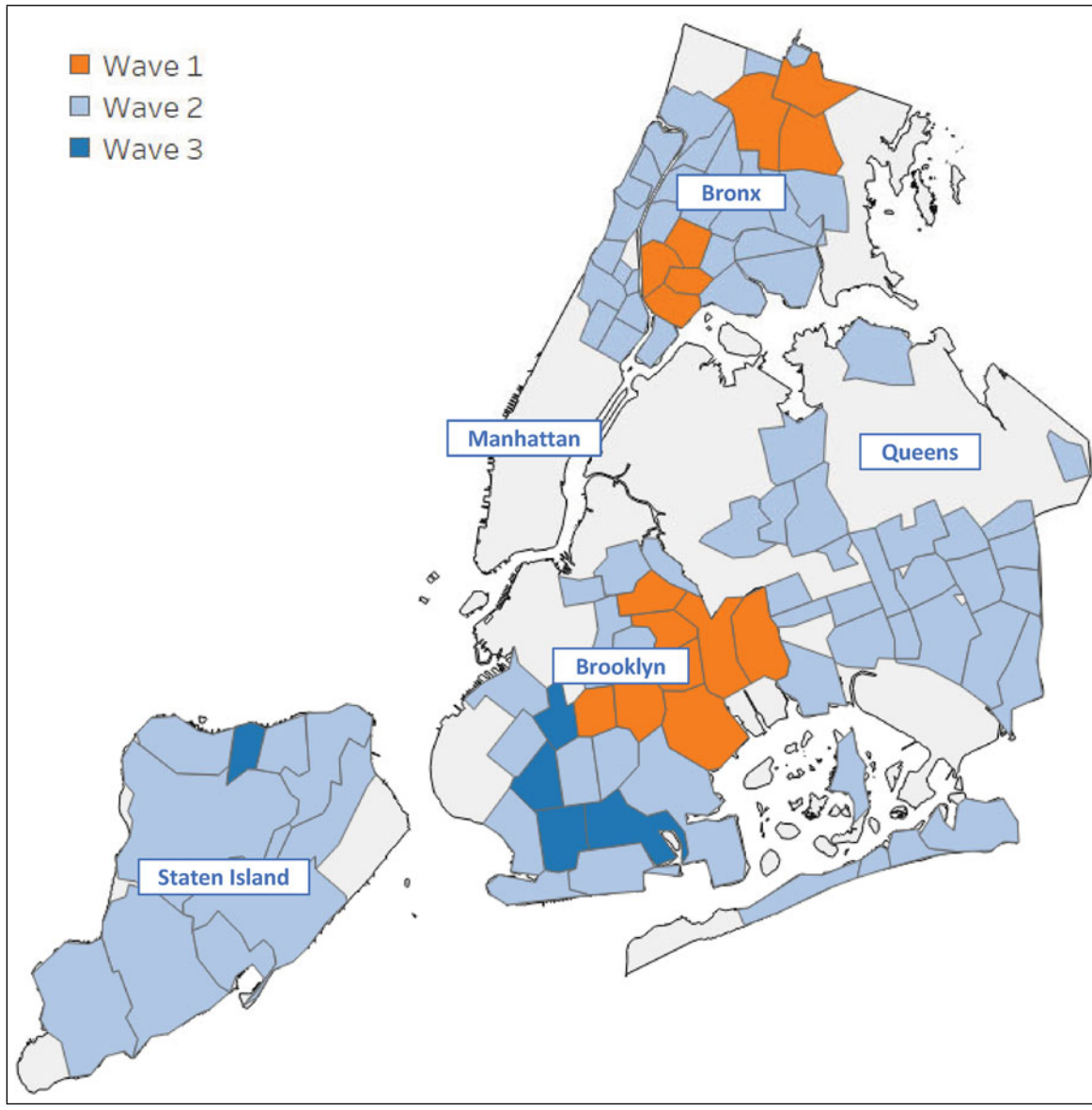


FIGURE Map of Targeted NYC Zip Codes
This figure is available in color online (www.JPHMP.com).

Evaluation

Program reach and assessment highlights

Through the 3 waves of VPHD, the team conducted 2873 detailing sessions (accounting for 2027 unique providers) at 1281 sites; detailing attempts were made at 2458 sites. Reps successfully completed detailing sessions at more than 85% of sites that were operating and in scope. Visits were generally well-received by providers and staff, and most were thankful for the individual-level outreach.

Data obtained through assessment questions, which were asked of each provider detailed, produced useful real-time feedback for DOHMH leadership. The assessment provided data on the following: perceived provider barriers to becoming a vaccination site; patient barriers to vaccination (as reported by providers based on communications with patients); awareness and implementation challenges of DOHMH initiatives; provider social media use; and resource and technical assistance needs. For example, it was an agency priority to enroll practices to serve as vaccination sites. Assessment data suggested that, over time, provider barriers to offering vaccination shifted from lack of storage/handling equipment for the vaccine and a perception that enrollment as a vaccine site was too burdensome to preferring to refer out for vaccination, given the wide availability of vaccine

throughout NYC. In contrast, assessment question data suggested patient barriers remained more consistent across neighborhoods and generally focused on a lack of trust in the vaccine development process, concerns about both immediate and long-term side effects of the vaccine, as well as a lack of perceived need for vaccination (typically because of previous COVID-19 diagnosis).

Another outcome of this campaign included referrals for technical assistance related to becoming a vaccination site or enrolling in the Citywide Immunization Registry. By design, technical assistance is not typically part of the detailing model, as visits are intended to be brief and educational in nature, while technical assistance can involve ongoing engagement. But consistent with DOHMH goals of increasing the number of vaccinating providers and improving vaccination administration reporting—and in response to requests for assistance from providers—VPHD offered to connect providers to other DOHMH programs for in-depth technical assistance. In total, 285 sites were referred for technical assistance during the campaign.

Another noteworthy finding was very few providers reported using social media as a way to gather public health information; most reported referring to government Web sites such as the Centers for Disease Control and Prevention or state and city health department Web sites. This finding was somewhat unexpected, as reports in the scientific and popular press suggested social media was a growing method for providers to stay abreast of rapidly changing COVID-19 best practices.[17-19] But VPHD reports suggest community providers continued to rely on more traditional means for public health updates.

Finally, because VPHD was intentionally delivered as one part of an extensive community reinvestment initiative, it is not possible to isolate the impact of VPHD alone on vaccination outcomes. Through the collective efforts of PACE initiatives, the equity gap in COVID-19 vaccinations was nearly eliminated: in December 2021, 91% of TRIE neighborhoods were considered "fully vaccinated neighborhoods" compared with 95% of non-TRIE neighborhoods (compared with 19% vs 64% in TRIE and non-TRIE neighborhoods, respectively, in July 2021) and 100% of TRIE neighborhoods were fully vaccinated neighborhoods in April 2023.[4]

Qualitative highlights

Qualitative information gathered by the VPHD team helped iteratively inform the DOHMH COVID-19 response. Reps gleaned qualitative information and "anecdata" while speaking with providers, contributing to a more holistic picture of COVID-19 vaccine sentiments across neighborhoods that was not captured through the assessment questions.

First, though the dynamic nature of the Action Kit materials prevented a formal assessment of their impact, reps collected information about providers' responses to the materials, which flowed back to the Public Information Officer to inform messaging. Providers reported the communication guides were especially useful, as many struggled to counsel vaccine-hesitant patients; other agency assessments similarly suggested misinformation materials helped medical staff feel better prepared to address and monitor the spread of misinformation.[20] With respect to VPHD, the key recommendations in later waves of the campaign addressed issues and questions raised by providers in earlier waves (eg, guidance against the use of serology testing to determine vaccination recommendations; see the Table).

Outside of information related to barriers to vaccination, several interesting trends emerged throughout the campaign, including:

1. Providers reported hesitancy to waste vaccine, even after DOHMH emphasized "Use Every Opportunity" guidance[21] and the associated concept of "useful wastage." Many felt what they described as an ethical responsibility to not open a vial for a single injection if the rest of the supply would go unused. This hesitancy translated into decreased provider/practice will to become a vaccination site or order more vaccine; this was particularly significant among providers with small panels or few unvaccinated patients.
2. Providers reported and reps saw misinformation around serology testing and the level of protection conferred by "natural immunity" spreading among both patients and providers. Providers reported patients were choosing not to get vaccinated because of a previous COVID-19 infection and assumed protection from existing antibodies. Reps also came across providers who were themselves purveyors of "natural immunity" misinformation, advising patients they did not necessarily recommend vaccination over "natural immunity" based on the data available. Some providers recommended using serology testing routinely (even monthly) to inform a patient's decision to get vaccinated on the basis of the patient's antibody levels; these providers tended to be clustered in communities with lower vaccination rates and misinformation often reflected community mistrust of the medical system. Reps also noted a wide range of antibody levels were being communicated as sufficient to

confer immunity for "robust" levels of protection (200-2500 units/mL).
3. Providers expressed a lack of confidence in either the vaccine or the promise of reaching high vaccination levels in the city due to perceived overpoliticization of the vaccine and vaccination behavior.

Discussion

VPHD served a critical and unique function as part of the agency's COVID-19 emergency response, helping bridge public health and health care delivery in a hands-on, high-touch way that centered racial equity and community reinvestment while planting seeds for future collaboration between the providers detailed and DOHMH. VPHD was also useful in tangible ways that speak to the needs of emergency response, including providing rapid feedback on agency materials and identifying resource needs. For instance, based on provider feedback reported via VPHD, the agency developed new guidance about how to speak with parents about COVID-19 vaccination in children to help address potential hesitancy.

In addition, providers often commented they were unaware of DOHMH resources or vaccination opportunities and programs, despite extensive public messaging. This feedback highlights the value of the in-person visit to not only provide information and resource dissemination but also offer encouragement and reassurance during a difficult time. The in-person visit was also uniquely able to address provider barriers specifically and offer support in a way public messaging or educational efforts such as webinars could not. By engaging in a one-on-one conversation, reps—who had deep familiarity with the agency resources—could offer a service-oriented approach and assist providers in overcoming barriers by quickly connecting the relevant materials with providers' challenges. Through their own work and in weekly check-ins with their colleagues, reps learned what types of strategies were successful in other primary care practices and how those practices utilized agency resources.

Finally, VPHD's placement within the PACE group led to amplified collective impact; while VPHD targeted medical providers, DOHMH's community engagement teams worked in the same neighborhoods to engage the general public. Both groups worked to educate, share resources, and gather community input with respective groups and the co-delivery of our outreach efforts provided opportunities for more collaboration within and reinvestment into historically marginalized communities. As noted earlier, these collective efforts significantly increased vaccination rates in targeted areas.[4]

Limitations and Challenges

One of the hardest aspects of the campaign was addressing rapidly changing scientific messaging. Utilizing tablets/e-materials facilitated sharing updated information, but word of mouth and misinformation often ran ahead of the science and DOHMH's ability to produce resources to counter misinformation. While VPHD worked to address misinformation as quickly as possible, it was an uphill battle to address the wide-ranging and swiftly evolving claims.

In addition, while the campaign was generally well-received, reps were occasionally met with feedback from providers who disagreed with aspects of NYC's COVID-19 response. As the campaign went on, commentary from providers became more politically charged and reflected the general societal landscape around vaccination, including skepticism about governmental efforts to promote vaccination or questions around vaccine efficacy. This type of feedback tended to be localized and reflected political leanings in the neighborhood. To address this, VPHD spent staff meetings training staff on neutral talking points or mechanisms for conflict de-escalation.

Another challenge was developing an accurate and up-to-date site list. While the agency works with community providers regularly, COVID-19 led to unprecedented changes in the health care landscape, with the closure of many small practices. Roughly 50% of the practices visited by reps were not viable because they were closed, not providing primary care,

Implications for Policy & Practice

- A provider outreach program such as PHD can be quickly adapted to address public health emergencies while prioritizing health equity and reinvestment in disproportionately impacted communities. Programmatic expertise in detailing and other communications strategies enables programs to rapidly mobilize.
- In-person outreach visits are well received by most providers and are an effective way to communicate urgent information to health care practices that are not reached by traditional public messaging. Relationships developed during detailing visits can also set the stage for ongoing relationships between community providers and local health departments.
- Strategic and purposeful collaboration between PHD and other community engagement initiatives can lead to amplified collective impact within communities.

or had policies in place preventing visitors due to risk of COVID-19 infection. To address this problem, local health departments could consider putting additional resources into developing and maintaining accurate provider contact lists during nonemergency times.

Conclusion

VPHD is an effective way to communicate with providers during a public health emergency while centering health equity. Among its advantages, the program can be scaled up or down as resources permit, thereby making it accessible for local health departments or organizations with limited resources. Its impact is further amplified by aligning with other community engagement programs occurring in the same neighborhoods. In addition, the program may assist with community building and establishing long-term relationships between providers and local health departments; if reps are viewed as trusted resources, these relationships can outlive a single campaign and forge ongoing opportunities for collaboration and bidirectional communication.

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